



Technoverse Hackathon 2026

Agent Builder Challenge
guidelines



Agent Builder Challenge guidelines

This document outlines the structure, expectations, and guidelines for the upcoming challenge. Please review the following instructions carefully.

01 Overview

The assessment will be conducted virtually on the Superset Platform. Ensure you have a stable internet connection and a suitable workspace before you begin, as the assessment cannot be paused once it has been initiated.

02 Agent Builder Challenge instructions

Duration: 45 Minutes | Mode: Team-Based

This round evaluates your team's ability to collaborate effectively, apply logical reasoning, and deliver structured solutions under time constraints.

Agent Builder is a workflow design challenge in which your team will construct a logical, node-based solution to a defined problem. You will be expected to design an end-to-end workflow, define data flow, and connect nodes to model your solution architecture.

Available node types include:

- Input, LLM Step, Prompt Template, Classifier/Router
- Condition/Branch, Tools, Knowledge Retrieval, Memory/Context, Output Formatter
- Evaluator/Guardrail, Retry/Fallback, Custom Code

This component assesses:

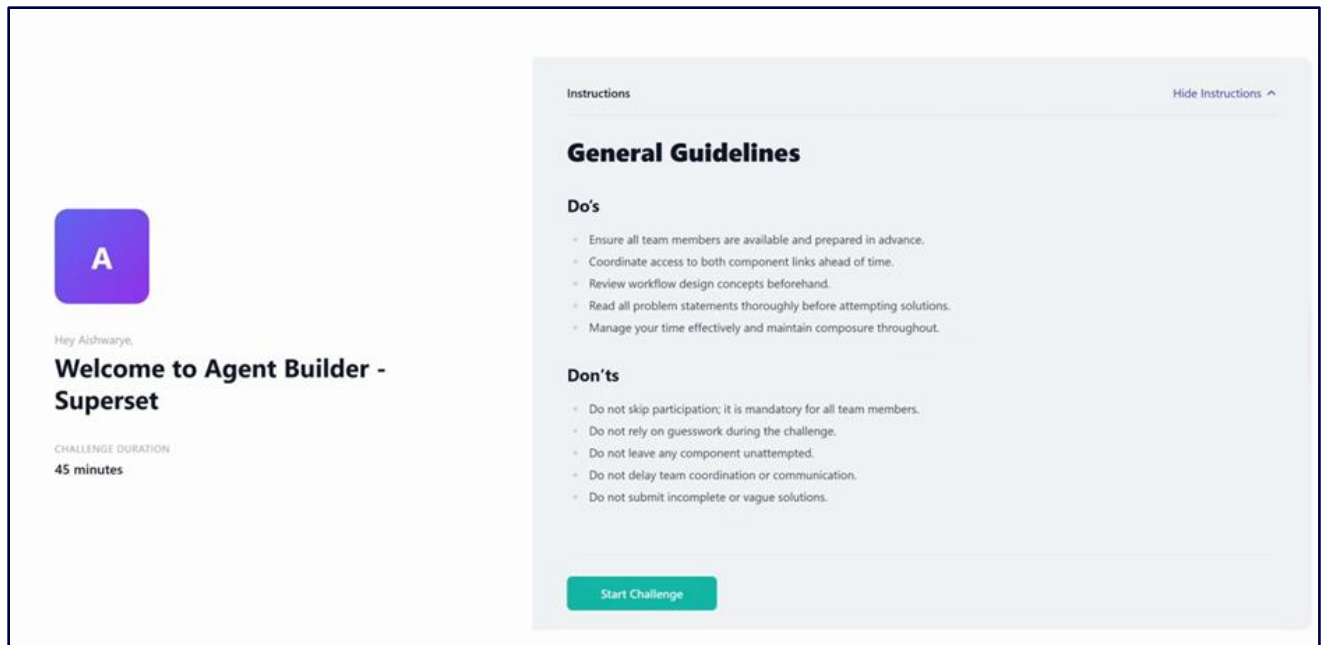
- Logical and structured thinking
- System design capability
- Clarity in solution architecture

03 Getting started with Agent Builder

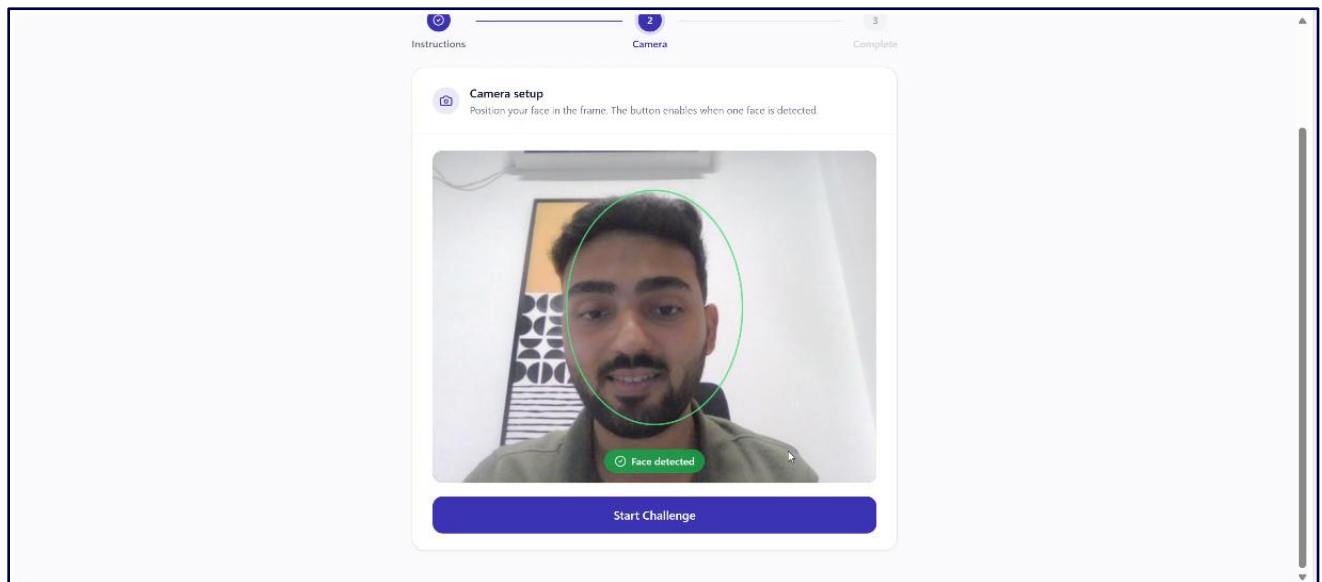
After successfully registering and forming your team, you will participate in the Agent Builder challenge. This section provides a walkthrough of the platform and the challenge process.

Initial Guidelines and Setup:

Upon entering the Agent Builder platform, you will be presented with a set of General Guidelines. The challenge duration is 45 minutes.



- It is mandatory to ensure all team members are available and prepared in advance.
- You must coordinate access, review workflow design concepts, and read all problem statements thoroughly.
- Do not skip participation, rely on guesswork, or submit incomplete solutions.



- Before starting the challenge, a camera setup is required for identity verification.
- Position your face in the frame; the "Start Challenge" button will enable once a face is detected.

04 Understanding the challenge

Once the challenge begins, you will be presented with the Challenge Brief.

For example, the "Retail Promotion Optimisation Agent" scenario requires designing a GenAI workflow to evaluate promotion effectiveness across various signals.

The screenshot displays the Superset web interface for a challenge titled "Retail Promotion Optimisation Agent". The interface is divided into several sections:

- Header:** Includes the Superset logo, a "Remaining 34:41" timer, and an "End Challenge" button.
- Scenario Section:**
 - SCENARIO:** A paragraph describing RetailCo's promotional campaigns and the inefficiency of current planning.
 - PROBLEM STATEMENT:** A box containing the task: "Design a GenAI workflow that receives a promotion planning input (product ID, channel, region, and basic context) and selectively calls relevant APIs to gather additional signals." It also includes success criteria: "Using the retrieved data, the workflow should: evaluate promotion effectiveness across demand, supply, and channel signals, use confidence-based logic to determine whether additional data is required, prioritise products and channels for promotion, determine the appropriate strategy (increase, maintain, reduce, avoid), and generate a clear rationale." Input details are listed: "Input includes product ID, channel, and region plus initial structured signals: - Current Promotion Level: High / Medium / Low / None - Inventory Status: Excess / Adequate / Limited / Stock risk".
- Start Building Section:**
 - A "Start Building" button with a right arrow.
 - AVAILABLE NODES (13):** A list of workflow nodes including Input, LLM Step, Knowledge Retrieval, Classifier Router, Condition Branch, Loop, Human Handoff, Output Formatter, Evaluator Guardrail, Custom Code, Prompt Template, Retry Fallback, and Merge Join.
 - REFLECTION QUESTIONS (5):** A section stating "After building, you'll answer 5 short questions about your design before submitting."

- The brief outlines the scenario, problem statement, and specific constraints, such as selective API usage and confidence-based branching.
- It also details the success criteria that your workflow must meet to be evaluated positively.



Remaining 34:37
End Challenge

the system recommendation engine, where any given request is evaluated for its characteristics, through a series of prompts and rules, and the system must show explicit confidence checkpoints. The workflow must show explicit confidence checkpoints.

- Supply-Demand Trade-off**
The system must balance promoting high-demand products to maximise revenue against avoiding promotion when supply is limited. Promoting a product with stock risk wastes budget and damages customer trust.
- Over-Promotion Risk**
High promotion frequency may lead to margin loss, brand dilution, and reduced incremental impact. The workflow must detect over-promotion signals and recommend reduction or avoidance where appropriate.
- Explainability**
Every recommendation must include a clear rationale: which APIs were called, why they were called, and how signals influenced the final promotion strategy and channel prioritisation.

SUCCESS CRITERIA

- ☒ APIs are called selectively based on input signals and intermediate confidence — not all APIs on every request.
- ☒ Promotion decisions are based on multiple signals (demand, supply, channel performance, competition, pricing sensitivity)
- ☒ Products and channels are prioritised logically — not all products receive the same strategy
- ☒ Promotion strategy (increase, maintain, reduce, avoid) is determined consistently for similar inputs
- ☒ Supply constraints are checked — limited stock does not receive increased promotion
- ☒ Over-promotion risk is detected and handled — high past frequency triggers reduce or avoid
- ☒ Trade-offs between competing signals are explicitly addressed (e.g., high demand but limited inventory)
- ☒ Unnecessary API calls are avoided
- ☒ A clear, structured rationale is generated for each decision including APIs used and signal reasoning

05 Building the workflow

- The core of the challenge involves using the Workflow Builder interface.
- On the left side of the screen, you will find the Node Palette, which contains all the available components you can use to construct your agent.



Remaining 34:37
End Challenge

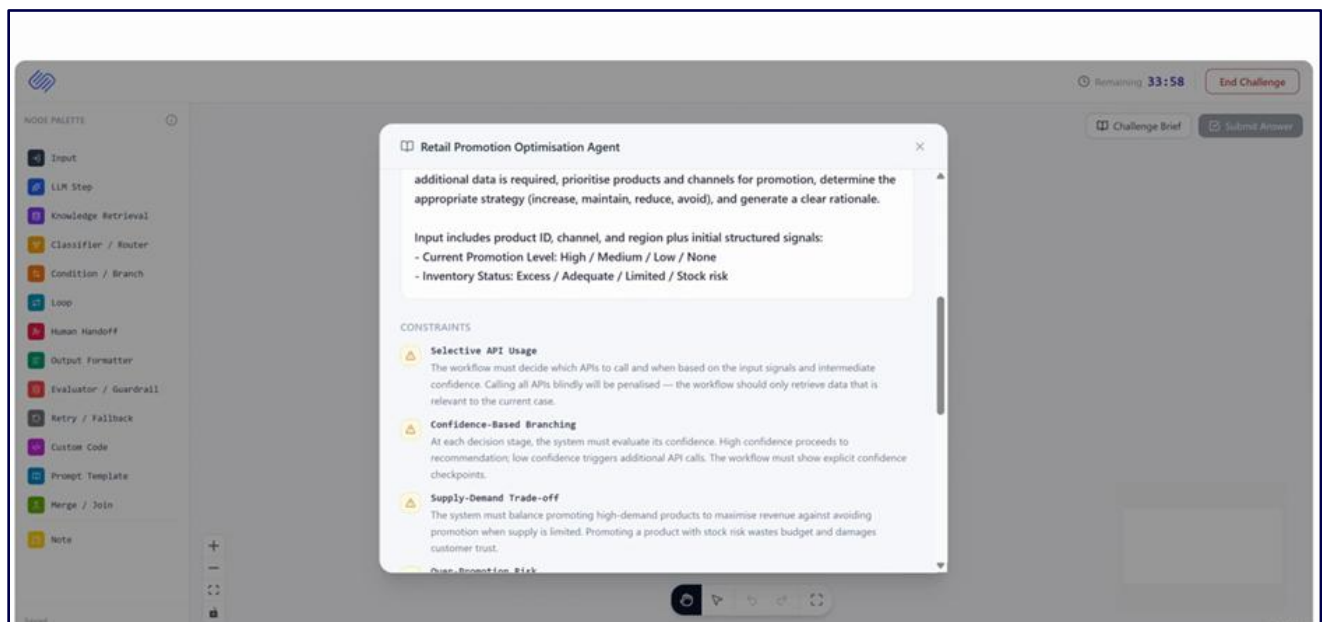
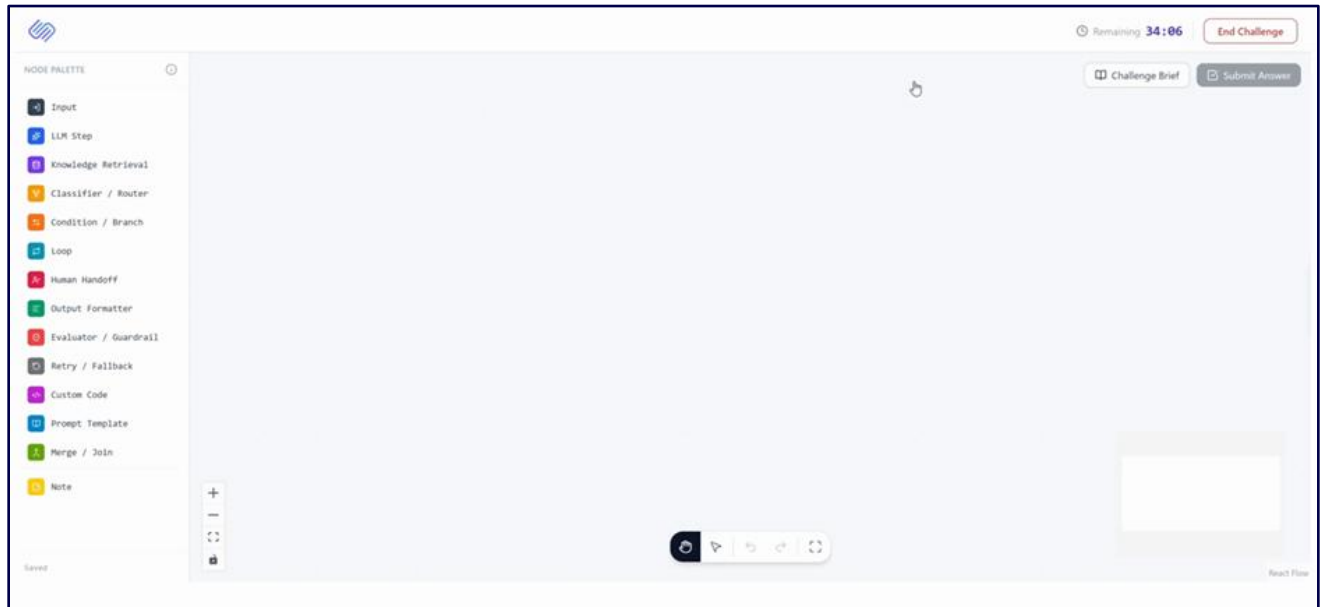
NODE PALETTE

- Input
- LLM Step
- Knowledge Retrieval
- Classifier / Router
- Condition / Branch
- Loop
- Human Handoff
- Output Formatter
- Evaluator / Guardrail
- Retry / Fallback
- Custom Code
- Prompt Template
- Merge / Join
- Note

Custom Code

Write a custom function to transform, filter, or process data.

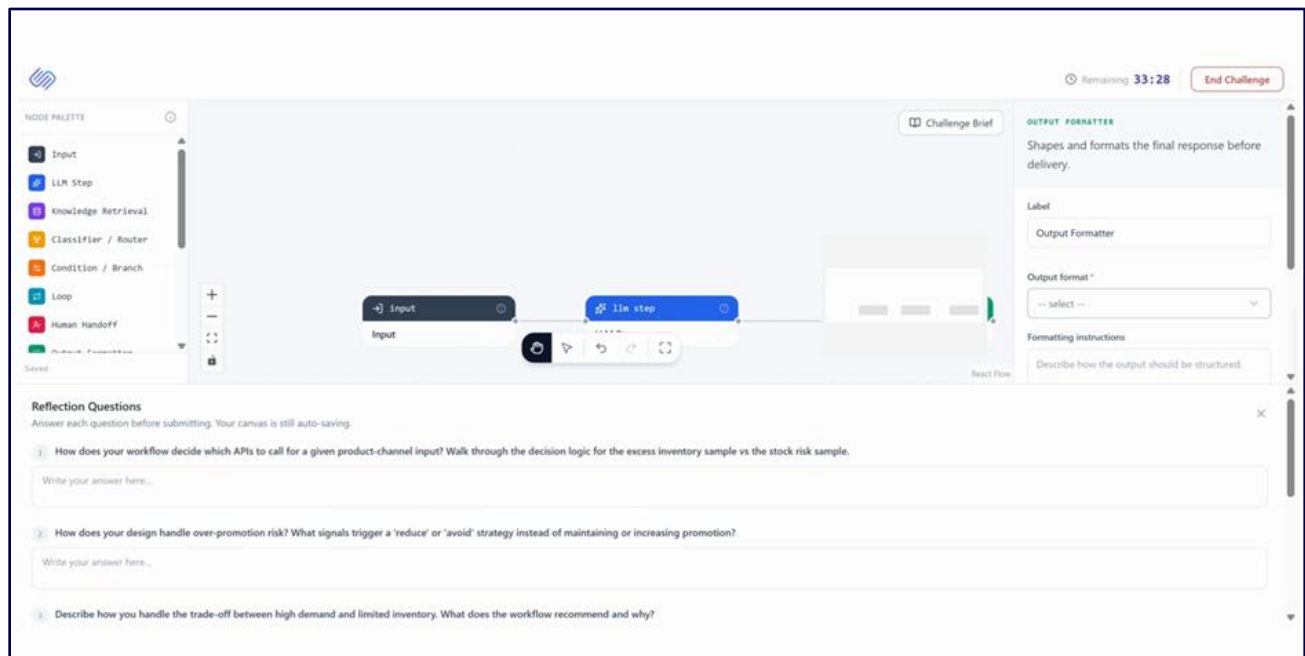
- Available nodes include Input, LLM Step, Knowledge Retrieval, Classifier/Router, Condition/Branch, Output Formatter, and Custom Code, among others.
- You drag and drop these nodes onto the central canvas to build your logic flow.



If you need to review the requirements at any point, you can click the "Challenge Brief" button to open a pop-up overlaying the canvas without losing your progress.

06 Final submission

After constructing your workflow, you must click the "Submit Answer" button. However, the submission is not immediately complete. You are required to answer a series of Reflection Questions based on your specific design.



- These questions ask you to explain your logic, such as how your workflow decides which APIs to call, how it handles over-promotion risk, and how it manages trade-offs between high demand and limited inventory.
- Providing clear, structured rationales is essential for a successful submission.
- Your progress is auto saved as you work on these questions.

In case of any queries during the Agent Builder Challenge (ABC), please reach out at <https://app.joinsuperset.com/support/#/>